

DNS Tunneling Detection System

Day 2 Data Ingestion, Domain Sanitization & Quality Validation Overview

PROJECT PHASE	RECORDS ANALYZED	TARGET PLATFORM	VALIDATION STATUS
Day 2: Ingestion & Prep	20,000 (15k Train + 5k Val)	Python 3.12 (Local Venv)	100% PASS

1. EXECUTIVE SUMMARY

Machine learning models are strictly bounded by data quality ("garbage in, garbage out"). In cybersecurity detection systems, raw network traffic often contains DNS protocol artifacts—such as the trailing root-dot delimiter specified in RFC 1034/1035 (e.g., `example.com.`)—and irregularities that can artificially skew feature engineering metrics.

The objective of **Day 2** was to ingest the raw, headerless datasets, implement a robust domain sanitization algorithm, enforce strict binary-label integrity, conduct comprehensive quality validations, and evaluate class balance and textual length characteristics before entering the feature extraction phase.

Day 2 Verdict: All 20,000 records across the training and validation sets were successfully ingested, cleanly sanitized, and validated with **zero missing values, zero empty strings**, and an exact **80% Tunnel / 20% Benign class distribution**.

2. DAY 2 MILESTONES & KEY DELIVERABLES

- Headerless Ingestion (`src/data_loader.py`):** Implemented `load_dataset()` to programmatically map Column 0 to `label` and Column 1 to `domain` without altering raw CSV files.
- Deterministic Sanitization (`src/preprocessing.py`):** Implemented `clean_domain()` to strip whitespace and remove trailing root-dots while strictly preserving internal subdomain dots.
- Integrity & Binary Validation:** Verified exact dataset dimensions (15,000 training, 5,000 validation) and confirmed all labels are strictly binary (0 or 1).
- Missing Values & Duplicate Audit:** Verified 0 missing labels, 0 missing domains, 0 empty strings, and audited duplicate domain records (28 in train, 8 in validation).
- Class Balance Confirmation:** Dynamically computed class counts, confirming both datasets reflect an expected 80% DNS Tunnel vs. 20% Benign domain distribution.
- Textual Length Profiling:** Evaluated minimum, maximum, mean, and median character lengths to establish a baseline for upcoming feature extraction.
- Updated Pipeline Orchestrator (`main.py`):** Integrated the end-to-end Day 2 verification pipeline with beginner-friendly, academic-grade terminal diagnostics.

3. INGESTED DATASET SPECIFICATIONS

Dataset	Total Records	Column 0 (Label)	Column 1 (Domain)	Format
training.csv	15,000 rows	0 = Benign, 1 = DNS Tunnel	Raw domain query string	Headerless CSV
validating.csv	5,000 rows	0 = Benign, 1 = DNS Tunnel	Raw domain query string	Headerless CSV

4. DOMAIN SANITIZATION LOGIC (SRC/PREPROCESSING.PY)

The `clean_domain()` function applies a 4-step deterministic transformation:

- 1. **Type Safety:** Safely casts non-null values to strings.
- 2. **Whitespace Stripping:** Strips leading and trailing spaces (`.strip()`).
- 3. **Whitespace Collapsing:** Normalizes irregular internal spacing (`" ".join(s.split())`).
- 4. **Trailing Root-Dot Stripping:** Removes trailing root-zone dot delimiters (`.rstrip(".")`). Internal dots separating labels are strictly preserved.

Verification Samples (Before vs. After Cleaning):

#	Raw Input (Before Cleaning)	Sanitized Output (After Cleaning)	Result
1	q+Z8AnwaBA.hidemyself.org.	q+Z8AnwaBA.hidemyself.org	CLEAN
2	q+Z8A3wbBA.hidemyself.org.	q+Z8A3wbBA.hidemyself.org	CLEAN
3	q+Z8BHwcBA.hidemyself.org.	q+Z8BHwcBA.hidemyself.org	CLEAN
4	q+Z8BXwdBA.hidemyself.org.	q+Z8BXwdBA.hidemyself.org	CLEAN
5	q+Z8BnweCORdAGL4+W7DB...hidemyself.org.	q+Z8BnweCORdAGL4+W7DB...hidemyself.org	CLEAN

5. DATA QUALITY & INTEGRITY AUDIT

Quality Dimension	Training Set (15,000)	Validation Set (5,000)	Academic Evaluation
Record Count	15,000 (Exact)	5,000 (Exact)	PASS Matches expected dimensions
Column Schema	['label', 'domain']	['label', 'domain']	PASS Standardized 2-column format
Ground-Truth Labels	Strictly {0, 1}	Strictly {0, 1}	PASS Zero corrupted/non-binary labels
Missing Labels	0 (0.00%)	0 (0.00%)	PASS No imputation needed
Missing Domains	0 (0.00%)	0 (0.00%)	PASS Complete domain records
Empty Strings	0 (0.00%)	0 (0.00%)	PASS No blank queries after cleaning
Duplicate Domains	28 records (0.19%)	8 records (0.16%)	Reported & preserved per Day 2 scope

6. CLASS DISTRIBUTION ANALYSIS

In real-world networks, attack traffic is often severely imbalanced. Here, both datasets exhibit a deliberate **80% Tunnel / 20% Benign** ratio:

Dataset	Total Records	Regular / Benign (0)	DNS Tunnel (1)	Ratio Match
Training Set	15,000	3,000 (20.0%)	12,000 (80.0%)	EXACT MATCH (80 / 20)
Validation Set	5,000	1,000 (20.0%)	4,000 (80.0%)	EXACT MATCH (80 / 20)

Academic Review Note: The identical 80/20 class proportions across training and validation ensure stratified, unbiased evaluation during Day 4 and Day 5 testing.

7. TEXTUAL DOMAIN LENGTH CHARACTERISTICS

Prior to feature extraction, basic length statistics confirm fundamental differences in payload structure:

Metric	Training Set	Validation Set	Cybersecurity Insight
Minimum Length	4 characters	4 characters	Shortest benign domains (e.g., t.co)
Maximum Length	245 characters	245 characters	Close to DNS RFC maximum limit (253 characters)
Average (Mean)	123.89 characters	124.18 characters	High mean reflects massive data exfiltration payloads
Median Length	137.0 characters	138.0 characters	Confirms that the majority of queries carry long encoded data

8. EXECUTION OUTPUT LOG (MAIN.PY)

```
=====
DNS TUNNELING DETECTION - DAY 2 DATA PIPELINE
=====
STEP 1: DATASET INGESTION
[Loader] Successfully loaded 'training.csv': 15,000 records.
[Loader] Successfully loaded 'validating.csv': 5,000 records.

STEP 2: DOMAIN SANITIZATION
[Preprocess] Sanitization completed for both training and validation sets.
Sample 1: q+Z8AnwaBA.hidemysself.org. -> q+Z8AnwaBA.hidemysself.org

STEP 3: DATA VALIDATION & INTEGRITY CHECKS
Training Set: 15,000 records | Labels=[0, 1] | Nulls=0 | Empty=0 -> [PASS]
Validation Set: 5,000 records | Labels=[0, 1] | Nulls=0 | Empty=0 -> [PASS]

STEP 4: CLASS DISTRIBUTION ANALYSIS
Training Set: 3,000 Benign (20.0%) | 12,000 Tunnel (80.0%) -> [MATCHES EXPECTED]
Validation Set: 1,000 Benign (20.0%) | 4,000 Tunnel (80.0%) -> [MATCHES EXPECTED]

STEP 5: BASIC DOMAIN-LENGTH STATISTICS
Training Set: Min=4 | Max=245 | Avg=123.89 | Median=137.0
Validation Set: Min=4 | Max=245 | Avg=124.18 | Median=138.0

=====
DAY 2 VERIFICATION VERDICT: [SUCCESS] ALL CHECKS PASSED!
=====
```

9. NEXT PHASE: ROADMAP TO DAY 3

With data ingestion and quality validation completed, the project is primed for Day 3:

- **Shannon Entropy (src/entropy.py):** Quantifying character randomness to identify Base32/Base64/Hex encoded tunneling payloads.

- **Lexical Feature Extraction (src/feature_extraction.py):** Extracting domain length, subdomain counts, digit ratios, and consonant-to-vowel ratios.
- **Feature Dataset Construction:** Assembling numerical feature matrices (X_{train}) and (X_{val}) ready for classifier training.